The Effect of Community Gardens on Neighbouring Property Values in London, a Difference in Differences Analysis

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1. Introduction

The construction of community gardens, often converted from vacant lots, has become a popular method for improving deprived urban environments. Numerous studies report positive economic, environmental and psychological benefits to their inclusion, which are pivotal for the counter arguments against their conversion to housing projects. Despite this, many of these studies rely on qualitative arguments and are predominantly centered in the American context, which limits the debates over whether to foster or replace community gardens. This is a critical gap in analysis, given the empirical value of community green space is central to debates over the desirability of greenspace policies and funding, the decision to charge developers impact fees for community gardens, and the result of qualitative studies are rarely transferable nor identify the direction of causality of any property value differences claimed by community gardens.

In this paper, we attempt to address these gaps using the Greater London area as our case study. It provides a particularly beneficial context given the richness of both new and historic allotments and is equipped with a swath of quality open-source data that has yet to be analysed for such purposes. A difference-in-differences (DiD) and event-study specification of a hedonic regression model is used to estimate the causal effect of community gardens on neighbourhood property values in the Greater London area from 2000-2025. Impacts are estimated based on the difference between property values in buffer zones around the garden's site before and after the year community gardens open and are in relation to the change of property value of comparable homes in the same LSOA. We examine the time needed for community gardens to begin effecting neighbourhood price, and how these effects vary with neighborhood depravity, garden size and location – i.e. the heterogeneity and homogeneity of the effect.

We find that, across a range of increasingly robust clustering methods, properties within 300 meters of a community garden experienced a 9.8% increase in sale price after the garden opened - in comparison to similar properties located between 300-1000 m from the same garden. Such trends begin to emerge around 8-10 years after opening, with peak coefficients typically found around 12 years after.

2. Literature Review

2.1 Community Gardens and Urban Sustainability

Existing literature on community gardens particularly highlights four key dimensions that would lead to property value increase. These dimensions are food security, ecosystem service, wellbeing and neighbourhood safety.

Caputo, Schoen and Blythe (2023) found that across 8 UK community gardens 16,979 kg of yields were produced in 2019, which translates to 42,447 people worth of daily fruit and vegetables per the UK governments recommended daily intake (NHS, 2022). A similar study by Sustain (2016) found that in 2013 2014 across 160 and 89 UK food gardens respectively the productivity was much smaller, 0.5 kg / m^2 versus 1.59 kg / m^2. Their study, although closer in representation to true portion productivity, also used more varied space types such as allotments and other small growing spaces.

Regarding environmental impacts, Lovell (2010) describes that as a result of shortened distance between production and consumer, packaging, pollution and water waste from urban agriculture is minimised in comparison to conventional agriculture. This result is contested as Goldstein et al. (2016) found that only some urban agriculture methods were more sustainable, however it varies significantly based on the climatic and resource context. A study by Speak et al. (2015) reported that, in allotment gardens from Poznan, Poland and Manchester, UK, across 26 ecosystem services including flood provision, climate regulation and pollination allotment gardens in both countries were greater contributors per unit area then parks.

Studies such as Davis, Green and Reed (2009) and Hartig et al. (2014) are among many which demonstrate the physically and emotionally dependency humans have on nature. As a result of the literature, it is widely recognised that spending time in nature improves wellbeing through stress relief and mental fatigue restoration, with gardening specifically being cited as a beneficial source of accomplishment (Kaplan, 1973). Such benefits of exposure to nature can be extrapolated to community gardening, which in itself contains literature suggesting the same benefits (Koay and Dillon, 2020). Amongst those that provide quantitative results, Wakefield et al. (2007) not only found similar improvements to mental and physical health as a result of the improved diets prompted by community gardening but specifically highlight the unique benefit of improved 'community health' as a result of improved relationships and a 'impetus for broader community mobilisation'.

Lastly, Bogar and Beyer (2016) did a systematic review of the relationship between greenspace and crime and violence. Their report found that 19 studies reported a positive impact whilst 9 reported a negative relationship. These differences were believed to result from differences in neighbourhood characteristics such as socioeconomic status, crime type and type of greenspace. In a more recent study, Koop-Monteiro (2021) used a fixed effects multilevel model to investigate the impact of new community gardens on neighbourhood crime in Vancouver, Canada. Their study reported a decrease in neighbourhood crime by 49 counts on average per garden added, empirically demonstrating evidence of suggested impacts of collective efficacy and crime resilience due to bonds formed and challenging narratives through local projects (Mccabe, 2014 and Jackson, 2017).

All in all, these four dimensions demonstrate how the inclusion of community gardens would improve the desirability of a neighbourhood, proven across numerous studies. Such improvements to a neighbourhood are likely to increase the price citizens would pay to live in them and hence motivates both the application of a DiD analysis and a grounding in hedonic pricing discussed more in section 2.3.

2.2 Community Gardens in London

The history of allotments and community gardens have deep roots in the Greater London area. Allotments first became available to the British public thanks to the Small Holdings and Allotment Act of 1907 (Bone, 1975). It then reached its highest during WWII as the 'Dig for Victory' campaign incentivised the conversion of vacant plots to allotments which produced ~10% of food consumed back then (Crouch and Ward, 1997 from Speak, Mizgajski and Borysiak, 2015). After the war, agricultural demand eased and the land was instead converted to houses and schools resulting in the decline of allotments from 1.5 million in 1950 to 250,000 in 2007 (Natural England, 2007). Despite the decline, around the turn of the millennium the National Society of Allotment and Leisure Gardeners (NSALG) saw a sudden increase in memberships and the reappearance of waiting lists demonstrating a spark for community gardens (Acton, 2011). The NSALG attributes the renewed interest to a Monty-Don television programme, the rollout of an Allotment Regeneration Initiative and growing government discourse of global warming and a weight gain crisis (Acton, 2011). Accordingly, many local authorities began to run schemes for regenerating and forming new community garden sites, leading to over 330,000 allotment gardens by 2023 and over 111,000 UK citizens on council waiting lists (Carhan, 2023). Whilst the preservation of the historic allotments from the early 1900s are largely incompatible with the current housing price data, the spike in popularity between 2004 to present justifies an analysis period following the year 2000.

2.3 Green Amenities and Housing Value

The literature examining the impact of community gardens on neighbouring house price only has one key study that sought to investigate the topic. Numerous studies have investigated the links between other green space types and property value, typically applying a variety of methods from census change to hedonic price modelling (Nicholls and Crompton, 2007, and Schinasi et al., 2021). These studies typically find a positive relationship between green space such as parks, lakes, churchyards and golf fields to house price, but also report a negative relationship with sites like industrial green buffers (Panduro and Veie, 2013). Such studies demonstrate that a relationship between green space and house prices likely exists, however the relationship is not homogenous and could be a property of systemic urban planning. In the sole direct community garden and house price study by Voicu and Been (2008), they also applied a DiD analysis investigating houses in New York neighbourhoods that either existed or were sold within 1000 ft of a community garden. The 1000 ft distance was arbitrarily chosen, however given community gardens are meant to serve the immediate area, it stands as a reasonable distance. Their model achieved a high R^2 of 0.86 and 0.78 for their residential and commercial models respectively, with statistically significant impacts found on properties within 1000 ft. More specifically, the coefficient of properties within 1000 ft of an existing or future garden was negative, selling for 11.1% less than similar properties outside the ring, whilst properties sold within 1000 ft of a garden had positive coefficients narrowing the gap by 3.6%. Their study demonstrates the effectiveness of applying a DiD methodology to the topic that produces causal evidence and limits the risk of conflating neighbourhood and time-invariant differences.

3. Methodology

3.1 Study Design and Data

The goal of the report was to apply a DiD and event-study specifications of a hedonic regression model to evaluate the impact of community gardens on neighbouring house prices across the Greater London area. To achieve this, a combination of datasets were used to collect site specific data across 17 gardens as well as land registry price data from the year 2000-2025. These sites and time span were selected on the basis of data availability, more specifically due to the fact that housing price data is restricted to 1995-2026 per HM Land Registry (2026) and only 17 verifiable community gardens were found to have opened after 1995 (see Figure 1). Based on the literature, we further suspect that community gardens began re-rising in popularity at the turn of the millennium, and thus our window is sufficient to observe both garden induced neighbourhood change and this specific shift in public perception. The opening dates data for each community garden was manually gathered based on reported dates for each garden published online and was merged on QGIS with the Ordnance Survey's (2025) open greenspace spatial dataset. The area of each garden was determined using the 'area' function on QGIS, and the English index of multiple deprivation (IMD) dataset from 2019 was merged to help control for broad neighbourhood socioeconomic conditions (Geographic Data Service, 2019).

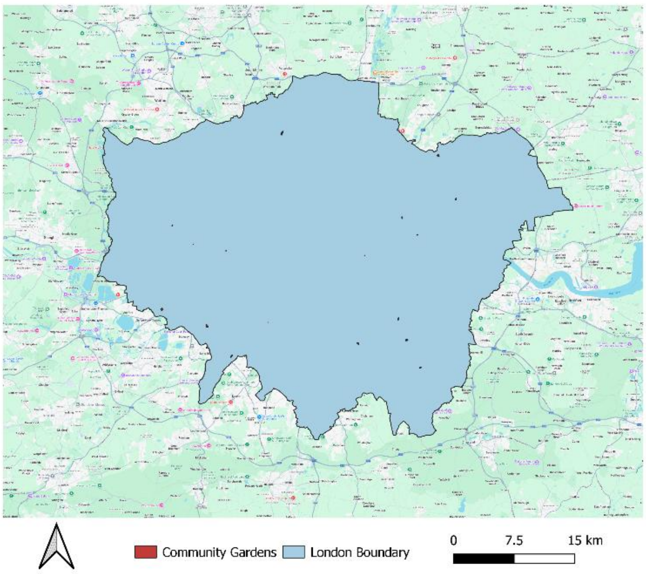

Figure 1. Map of community gardens chosen

3.2 Processing Pipeline

To merge the datasets and produce the DiD and event-study analysis a comprehensive pipeline was constructed in R as shown in figure 2.

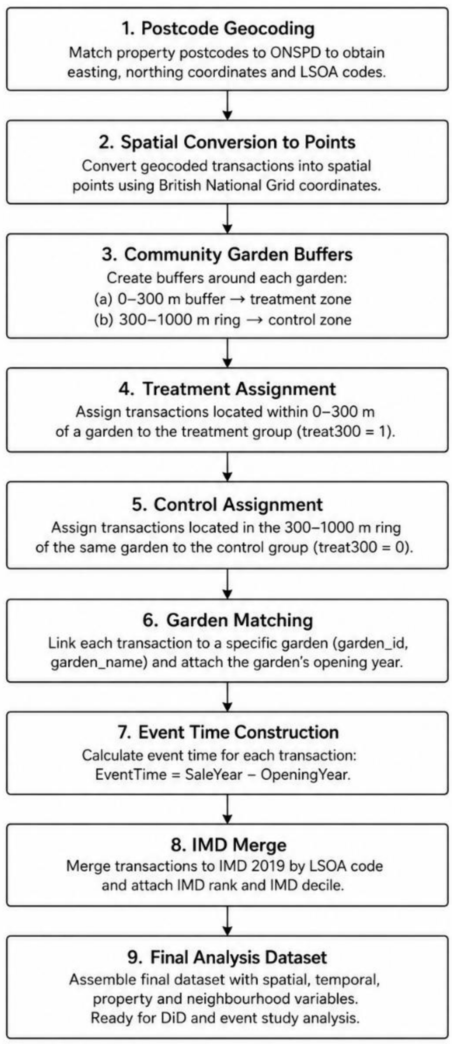

Figure 2. Workflow diagram of R processing pipeline for final dataset

The first step was to geocode the property transaction data as the original dataset does not provide coordinate locations. To accomplish this, transactions were filtered to Greater London based on their area key and then merged with the national postcode directory which provides easting and northings based on the postcode (Office for National Statistics, 2025). This serves to both filter transactions without matching coordinates and additionally merges LSOA identifier labels into the dataset.

With the transaction point data generated, each property is then assigned to the nearest community garden based on whether they intersect with either the 300 m treatment buffer or 300-1000 m control buffers created around each imported community garden. The 300-1000 m buffer serves as a control factor to capture the housing surrounding the neighbourhood of a community garden, meanwhile the more discrete 300 m buffer was used to define the treatment area – as chosen in Voicu and Been (2008). In the dataset, they are labelled treat300 and control and are binary dummy-variables that indicate whether a property is within a treatment and-or control buffer. The exact distance chosen was semi-arbitrary serving the purpose of reducing the comparison from city-wide to similar near-neighbourhood housing to reduce the bias from neighbourhood heterogeneity.

Following the garden matching, the IMD 2019 dataset is merged to each property using their LSOA code. The 2019 dataset was chosen as it was the most up-to-date dataset still within approximately 5-10 years of each community garden opening. This is key as using an earlier dataset from the late 2000s to early 2010s could potentially misrepresent improved neighbourhoods as the literature predicts, but using the newest 2025 rankings would be too far detached from the opening of most 21^{st} century community gardens.

The final step was the event time construction, which follows applying the simple equation shown below to each transaction to find the time difference between the property sale and the opening year of the community garden.

$$EventTime_{it} = SaleYear_t - OpeningYear_g$$

Eq 1. Event time construction, where t is the property transaction and g is the assigned garden

In total, 134,055 property transactions are observed and assigned to one of the seventeen community gardens investigated.

3.3 Difference-in-Differences

The application of a DiD model allowed us to investigate whether properties located near community gardens experienced a different change in sale price after garden establishment compared with similar properties located further away. It achieves this by comparing how prices changed before and after establishment between treatment and control areas whilst factoring for fixed effects such as neighbourhood quality and city-wide changes over time. Our model follows a typical ordinary least square (OLS) DiD regression shown in equation 2 but is modified to fit for our specification as shown in equation 3.

$$Y_{it} = \alpha + \beta_1 Post_t + \beta_2 Treat_i + \beta_3 (Treat_i \times Post_t) + \varepsilon_{it}$$

Eq 2. DiD implementation with OLS and two-way fixed effects. In our case, Post is 0/1 if the sale occurs before/after the garden opened, Treat is 0/1 if the properties lies within/outside 300 m of a community garden and Treat * Post is our DiD interaction

$$\ln(P_{igt}) = \beta_0 + \beta_1 Treat_{ig} + \beta_2 Post_{gt} + \beta_3 (Treat_{ig} \times Post_{gt}) + \gamma X_{it} + \delta IMD_i + \mu_g + \lambda_t + \varepsilon_{igt}$$

Eq 3. DiD implementation for the specifications of our chosen final model where ln(P) is the natural log of the sale price, X is property characteristics, IMD is our neighbourhood deprivation measure, μ_g is our garden fixed effects and λ_t is our year fixed effects

As shown, our coefficient of interest is β_3 which measures the average treatment effect of community garden on house sales in 300 meters. Additionally, we include a fixed effects and year effects term to help factor heterogeneity between neighbourhoods, gardens and yearly economic trends (i.e. inflation, policy and macroeconomic shocks). Lastly, we also include a property characteristics term which includes information on whether the property is a detached and freehold property as these conditions typically lead to higher sale prices.

3.4 Event-study Model

The DiD model follows a key assumption called parallel trends outlined in equation 4. The interpretation of this assumption, in our context, believes that in the absence of a community garden it is expected that treated and control properties in a neighbourhood would have followed similar changes in price.

$$E[\Delta Y(0) \mid Treat] = E[\Delta Y(0) \mid Control]$$

Eq 4. Parallel trends assumption

Whilst a DiD helps determine the average treatment effect; by conducting an event-study we can evaluate the validity of the parallel trends assumption by investigating pre-treatment trends, treatment effect timings and the persistence of the treatment's impact. Our event-study follows the formula outlined in equation 5.

$$ln(P_{igt}) = \sum_{k \neq -1} \beta_k \left(Treat_i \times 1[event_{time} = k]\right) + \gamma X_i + \delta IMD_i + \mu_g + \lambda_t + \varepsilon_{igt}$$

Eq 5. event-study model, where k is the specific event time i.e. -15, … ,15, and k = -1 is omitted as the reference period

Each β_k represents the price difference between treated and control property transactions k years from when the garden was built – relative to one year prior to opening. By plotting such results, the model can verify if the parallel trends assumption is observed, the rate at which the treatment begins to change transaction prices compared to the control (i.e. immediate, gradual or delayed), or the potential for anticipation effects due to endogenous garden placement. As outlined in Roth (2024), 'an event-study plot is more convincing evidence in favour of a non-zero treatment effect if there is a discontinuity in the coefficients close to the treatment date'.

3.5 Clustering Robustness

An important consideration when constructing the DiD and event-study models is that the default relationship assumes i.i.d property transactions. In this case, it is highly unlikely given planned local policy, neighbourhood characteristics and environmental conditions that properties sales around the same community garden are not spatially correlated and thus contribute non-independent information. By choosing to ignore such characteristics it would cause the model to underestimate standard error and inflated statistical significance. This can be counteracted by clustering standard error based on the expected correlations and in our analysis is achieved by clustering at the garden, LSOA and two-way (garden & LSOA) level.

4. Results

4.1 Difference in Differences

Table 1 summarises the results of improving on an initial DiD regression of just the treatment and control factors to a full hedonic price DiD regression. The key finding is that, as additional control variables were added, the interaction coefficient increases from 0.067 to 0.093, or an average increase of 6.9% to 9.8% in transaction value for sites near community garden openings. These results are statistically significant against a confidence interval of 99% ($p < 0.01$) across each model and also show that IMD is largely unimpactful on model estimates whilst building characteristics contributed to a significant improvement. This can be seen by the change in R^2 values, which grew from 0.47 to 0.63 and more specifically the within R^2 value that jumped from 0.001 to 0.304. Within R^2 removes the impact of garden and year fixed effects thus demonstrating that the interaction term had little explanatory power in the baseline, but after adding tenure and property type was explaining ~30% of the variation.

Variables	Baseline DiD	DiD + IMD	Full Hedonic DiD
Treatment Area (300m)	-0.0358.	-0.0313	-0.0628***
	(0.0204)	(0.0236)	(0.0123)
Post-Opening Period	-0.0321	-0.0280	-0.0381
	(0.0212)	(0.0204)	(0.0227)
Treatment × Post	0.0665**	0.0807**	0.0934**
	(0.0171)	(0.0244)	(0.0296)
IMD Decile		0.0561***	0.0269**
		(0.0094)	(0.0082)
Flat			-0.5406***
			(0.0423)
Other			-0.9721***
			(0.1442)

Semi-Detached			-0.4203***
			(0.0498)
Terraced			-0.6011***
			(0.0573)
Leasehold			-0.4914***
			(0.0463)
Unknown Duration			0.0515
			(0.0428)
R^2	0.471	0.487	0.631
Within R^2	0.001	0.031	0.304

Table 1. DiD model sophistication expansion. Note all models include garden and year effects, standard errors are in parentheses, all models are clustered by two-way garden and LSOA and significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 2 summarises the results of specifying the standard error clustering in the DiD regression from i.i.d to two-way garden and LSOA clustering. The key finding is that as clustering becomes more specific the treatment-control interaction term remains statistically significant against a confidence interval of 99% ($p < 0.01$). There is clear evidence of standard estimation deflation as the interaction term standard error increases from 0.004 in i.i.d to 0.023 in two-way clustering - indicating the presence of collinearity amongst transactions. Even still, the robustness of statistical significance across clustering suggests that the estimated house price premium is primarily driven by the interaction term, fixed effects and building characteristics, and not the choice of variance estimator.

Variables	IID	Garden Cluster	LSOA Cluster	Two-Way Cluster
Treatment Area (300m)	-0.0628***	-0.0628***	-0.0628***	-0.0628***
	(0.0044)	(0.0123)	(0.0139)	(0.0123)
Post-Opening Period	-0.0381***	-0.0381	-0.0381*	-0.0381
	(0.0044)	(0.0227)	(0.0171)	(0.0227)
Treatment × Post	0.0934***	0.0934**	0.0934**	0.0934**
	(0.0072)	(0.0296)	(0.0328)	(0.0296)
IMD Decile	0.0269***	0.0269**	0.0269***	0.0269**
	(0.0008)	(0.0082)	(0.0055)	(0.0082)
Flat	-0.5406***	-0.5406***	-0.5406***	-0.5406***
Other	-0.9721***	-0.9721***	-0.9721***	-0.9721***
Semi-Detached	-0.4203***	-0.4203***	-0.4203***	-0.4203***

Terraced	-0.6011***	-0.6011***	-0.6011***	-0.6011***
Leasehold	-0.4914***	-0.4914***	-0.4914***	-0.4914***
Unknown Duration	0.0515	0.0515	0.0515	0.0515
R^2	0.631	0.631	0.631	0.631
Within R^2	0.304	0.304	0.304	0.304

Table 2. DiD clustering robustness. Note all models include garden and year effects, standard errors are in parentheses and significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

4.2 Event-study

Figure 3 summarises the results of the event-time study with two-way standard error clustering. The key findings are that, firstly, the parallel trends assumption is supported. Prior to $k = 0$, most coefficients lie close to and have confidence intervals that overlap with zero ranging from -0.1 to -0.05. Two outlier prior points exist at $k = -23$ and 24 however these points themselves suffer from higher noise which typically occurs in event-studies over long horizons. Secondly, the effect of treatment does not begin to differentiate from the control until approximately 10 years after treatment occurs. At $k = 10$, there is an increase of 0.05 log points, with the peak increase found between 12-15 years after at ~0.15-0.2 log points growth. This indicates a lagged and dynamic effect of community garden implementation on housing. Lastly, after 15 years confidence intervals become less stable and coefficients return to zero, which could be a result of either exogenous effects and or shrinking sample size from fewer gardens at the ends of the horizon.

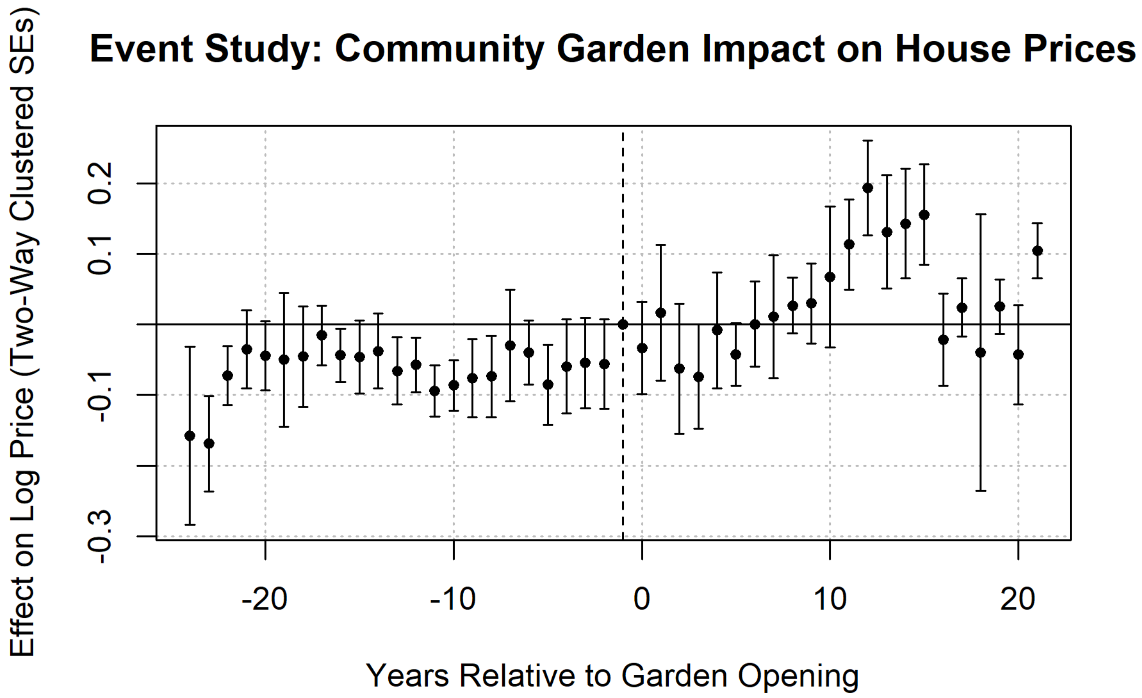

Fig 3. Event-time study of community garden impact on house prices clustered using two-way clustering (other cluster types in Appendix 1).

5. Discussion

5.1 Interpretation and Policy Implications

The key overall finding is that community gardens were found to have on average a 9.8% increase on transaction value in property within 300 meters, with this effect typically seen 8-10 years after opening. This mimics the results found in Voicu and Been (2008), who found an initial 6.2 percent increase in treated transactions after gardens opened in New York - with future effects growing by up to 9.4% five years after. This ultimately supports the main argument to protect community gardens over redevelopment projects as their inclusion in both Greater London and New York are found to contribute greatly to neighbourhood value. That said, there also exists a clear delay in such returns, particularly in London, which without secure political protection could easily have community gardens dismissed early on. The potential explanations for this are, as Voicu and Been (2008) highlight, community gardens particularly benefit lower-income neighbourhoods and could thus require some reputation transformation to increase neighbourhood value. In our study, this was partially supported by the significant explanatory power accounted for by the fixed effects term that indicate strong neighbourhood influences and is explicitly tested and shown in Voicu and Been (2008). Logically, this also follows based on the fact that community gardens require time for membership to grow, agricultural outputs to develop and bear fruit and the community bond and crime resilience highlighted by Mccabe (2014) to establish itself. In addition to protecting the establishment of these sites for periods of up to 8 years, its likely that any promotional and supportive efforts to encourage and guide users how to best utilise the space will help accelerate the benefits of the garden and hence increase site value. There is also a discussion to be had about the ethics of such improvements. The literature already demonstrates that any interactive greenspace is likely to increase neighbourhood prices, and, given community gardens likely rely on the build-up of community bonds, such efforts would be futile if the initial neighbourhood is flushed out by rising prices.

5.2 Limitations

The investigation, although in support of the claim that community gardens quantitively increase neighbourhood property value, suffers from key limitations that inhibit the scale of interpretation. Firstly, only 17 gardens were analysed, which are unlikely to be representative of the different neighbourhood and spatial patterns in the Greater London area. This is further compromised by the non-random placement of gardens, which also given the clustering methods, mean the impact of a low sample size on inflated positive results increases. As such, it is difficult to claim based on these results that the 9.8% mean housing premium would apply across the city but given the methodology it is fairly certain that this applies to the sample chosen that particularly concentrates around outskirt areas. This also opens a potential extension to verify the finding of 17 newly opened community gardens following the year 2000. Perhaps through remote sensing or independent council investigations the true figure could be established.

Another key issue is to do with limited control variables breadth. The IMD fixed at 2019 is not a dynamic measure of changing deprivation over the study period, nor does it help represent other regeneration projects that are likely to coincide with garden openings. Additional variables to include would be those that measure the quality of the different gardens, perhaps based on membership or output, and precise neighbourhood socio-economic characteristics that could differentiate impact on lower and higher income neighborhoods.

6. Conclusion

To conclude, the report aimed at applying a DiD methodology to quantitatively measure the impact of community gardens on similar neighbouring property transactions from 2000-2025. Our findings conclude that the inclusion of community garden increases property value by 9.8% on average; with such effects typically triggering significantly after its establishment. Evidence of pre-trends was minimal and robust across clustering methods demonstrating both the effective choice of a DiD methodology and non-i.i.d transactions. The findings help to justify the argument for the protection of community gardens but also emphasise that the value is likely derived from community empowerment and thus needs to be nurtured and given appropriate time. Future studies should aim to increase the number of gardens analysed and factor in the quality of the gardens to further validate the causality and application of the findings across the Greater London area.

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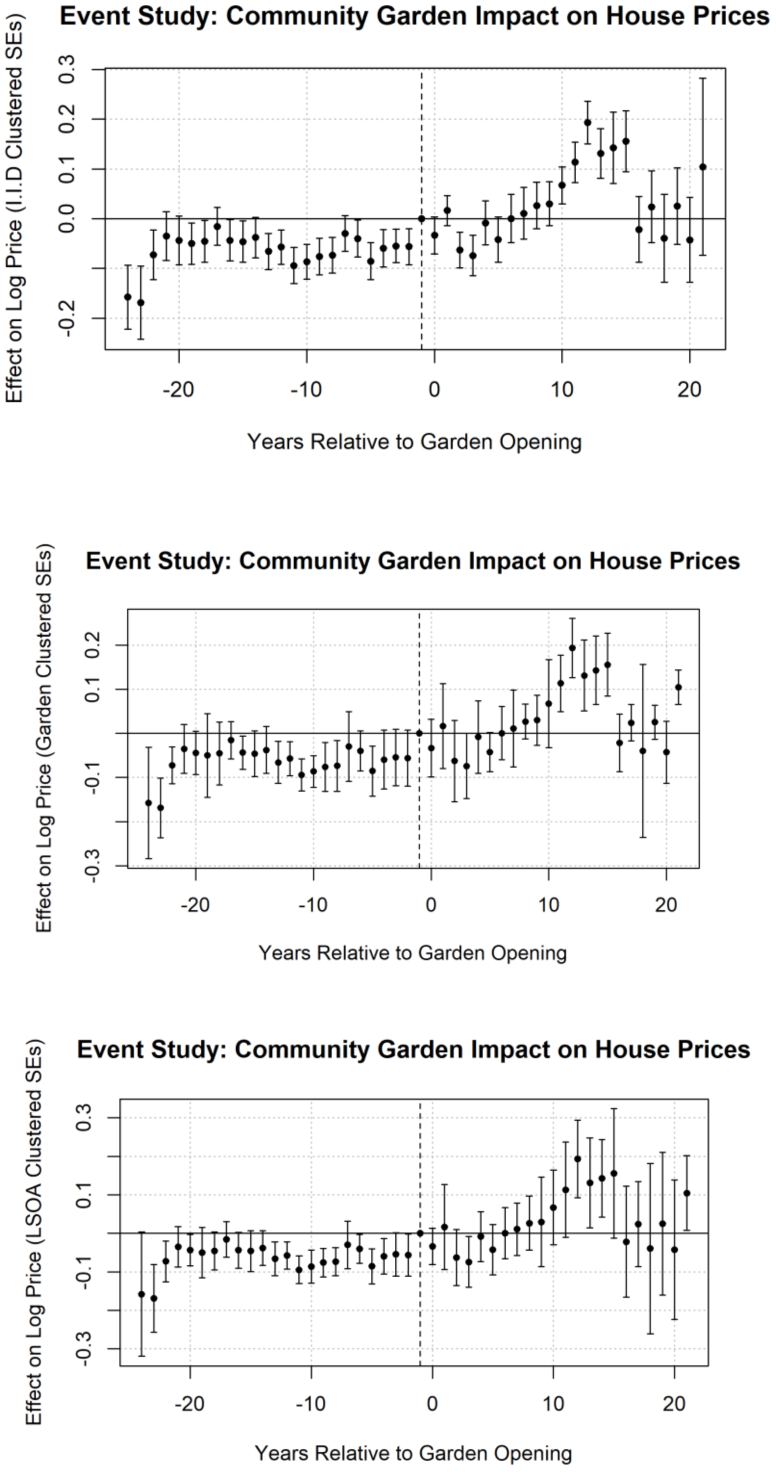
8. Appendix



Appendix 1: Event-time study of community garden impact on house prices clustered using i.i.d, garden ID and LSOA